

5. Show that given any rational number x , there exists an integer y such that x^2y is an integer.

Hint: Since x is rational, we can write $x = \frac{m}{n}$ for some integers m and n , where $n \neq 0$. All you need to do is to describe y in terms of m and n .

$$x = \frac{m}{n} \quad n \neq 0 \quad m \text{ and } n \text{ are integers}$$

$$\text{prove : } \left(\frac{m}{n}\right)^2 y \in \mathbb{Z}$$

$$\frac{m^2}{n^2} y \in \mathbb{Z}$$

so m^2y needs to be
divided by n^2

$$\text{lets say } y = n^2$$

$$\frac{m^2 n^2}{n^2} = m^2 \in \mathbb{Z}$$

since m^2 is an integer
it passes the condition

$$\boxed{\text{proving } x^2 y \in \mathbb{Z}}$$